

EFY Note

The source code of this project is available for free download at source.efymag.com

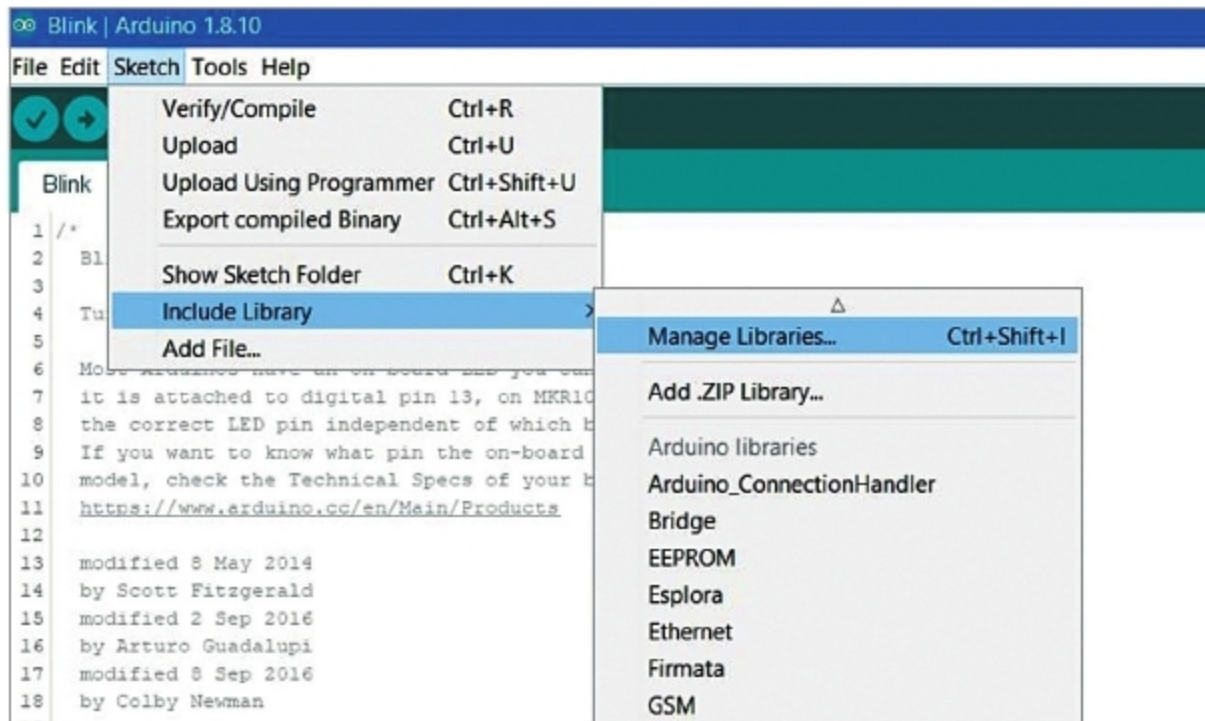


Fig. 4: Manager Libraries in Arduino IDE

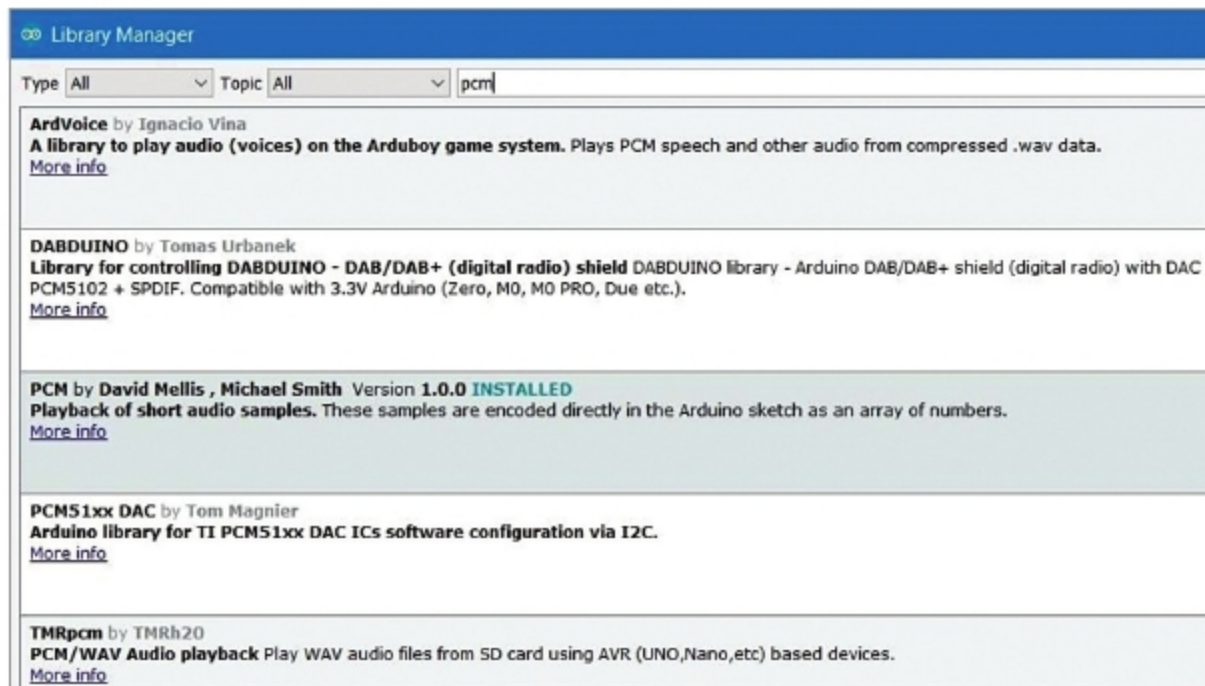


Fig. 5: Search PCM library from Library Manager

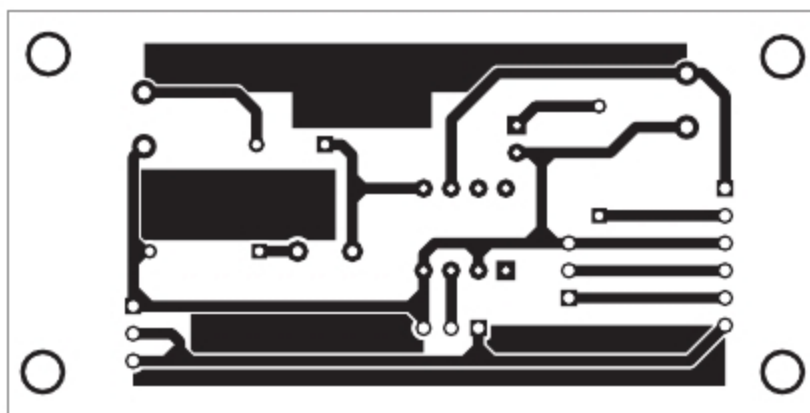


Fig. 6: Actual-size PCB layout for motion sensor alarm

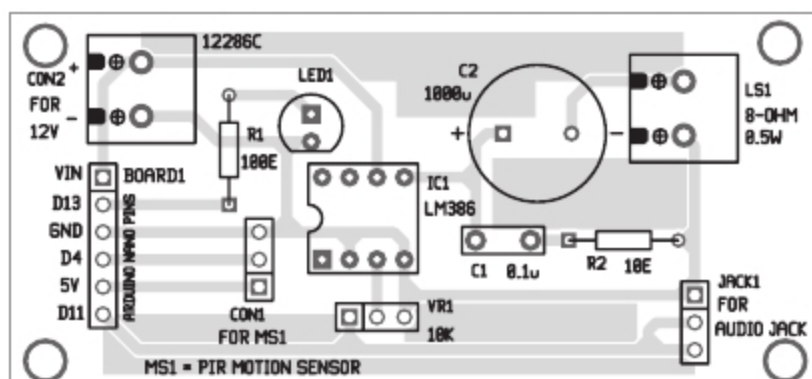


Fig. 7: Components layout of the PCB

```
if (sensorValue == 1)
{
  startPlayback(sample,
    sizeof(sample));
  digitalWrite(led,HIGH);
}
else if (sensorValue == 0)
{
  digitalWrite(led,LOW);
}
```

Construction and testing

An actual-size PCB layout for the motion sensor alarm is shown in Fig. 6 and its components layout in Fig. 7.

Before assembling the circuit, do not forget to upload Motion_Detection.ino on the Arduino

Nano board. After assembling the circuit on the PCB, connect the PIR sensor across CON1 and 12V across CON2.

Crosscheck the wiring/connections as per the circuit diagram. Then, power on the circuit with a 12V DC supply or 12V battery.

You can install this project in your backyard or any other appropriate location in your house. Then extend the speaker and wires to your room. Alternatively, you can connect it to your mini speaker amplifier through an audio jack.

When you power on the circuit, and if someone is in front of the PIR sensor, you will hear 'alert motion' voice alert in the speaker continuously till the person moves away from the sensor. This sensor will detect a human in the range of up to ten metres based on the sensitivity of the sensor module. You can change the sensitivity by adjusting the inbuilt potmeter in the sensor module. **EFY**



Venkatesh T. is an electronics hobbyist. His areas of interest include microcontrollers, embedded systems and the Internet of Things